

High Sales Don't Always Mean High Profit (The Furniture Story)

Nurhuda Dwi Utami



About Me

HELLO,
I'M TAMI

Data Analyst & Business Intelligence Enthusiast and **Certified Business Intelligence Analyst** with a passion for turning complex data into meaningful insights that drive smarter decisions. **Skilled in SQL, Google Sheets, Microsoft Excel, Tableau, and Looker Studio**, with strong experience in data processing, dashboard development, and daily performance reporting.

Comfortable working on end-to-end analysis, from data cleaning and transformation to visualization and storytelling, ensuring every number translates into a clear business narrative. Known for combining analytical precision with creative storytelling to deliver insights that inspire action and impact.

Technical Skills

PROGRAMMING LANGUAGE



DATA ANALYTIC TOOLS



Microsoft Excel



Google Sheet



Google BigQuery

DATA VISUALIZATION TOOLS



Tableau



Looker Studio

DATA ANALYTIC METHODS

- EDA
- Segmentation/Clustering
- Linear Regression
- Logistic Regression
- Statistics
- ANOVA
- t-Test

Education & Experience Background

EDUCATION



DQLab

Bootcamp Data Analyst with SQL & Python in Google Platform (2025)

Attended the mini course and intensive data analytics course with Python and SQL.



Mercu Buana University

Master of Management

Cumulative GPA : 3.83 / 4.00



Islamic University of Indonesia

Bachelor of Statistics (2017)

Cumulative GPA : 3.82 / 4.00

EXPERIENCE



Ministry of Public Works

Planning Officer (2025-Present)

Engaged in organizational risk management, integrity zone implementation, and internal compliance activities.



Ministry of Trade

Statistician (2023-2025)

- Conducted statistical analysis to identify export opportunities and high-potential destination countries.
- Prepared monthly export–import performance reports to support trade policy and decision-making.



National Public Procurement Agency

Data Analyst (2019-2023)

Conducted data analysis for monitoring and evaluation of government procurement performance across Indonesia

EXECUTIVE SUMMARY

1 Project Background

- Profitability across product categories shows inconsistent performance.
- Furniture category records significant profit decline when discounts exceed 30%.
- Need to understand how sales, discount, and product category affect overall profit performance.

2 Insight and Recommendation

- Limit discount range and adopt data-driven pricing strategies to protect margins.
- Prioritize Technology and Office Supplies for marketing and stock allocation.
- Re-evaluate Furniture pricing model to reduce over-discounting and improve category profitability.

3 Regression Analysis & Key Findings

- Linear regression shows Discount has a strong negative and significant effect on Profit ($B = -1,520,872$, $p < 0.001$).
- Sales has a positive and significant relationship with Profit ($B = 0.228$, $p < 0.001$).
- Compared to Furniture:
- Technology yields +450K higher profit.
- Office Supplies yields +125K higher profit.
- The model confirms that Furniture remains the least profitable category.

4 Impact Analysis

- High discounting erodes profit margins, especially in Furniture.
- Even at similar sales volume, excessive discounts can reduce profit by up to 7× compared to Technology.
- Controlling discount rates could significantly improve overall margin performance.

METHODOLOGY

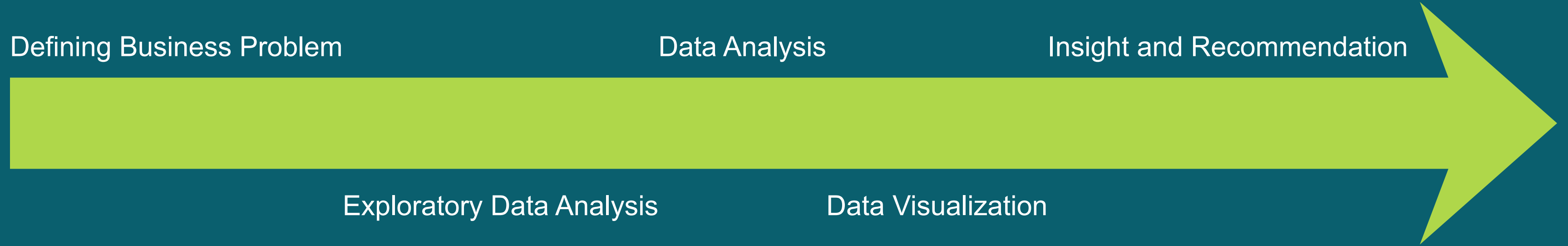
Defining Business Problem

Data Analysis

Insight and Recommendation

Exploratory Data Analysis

Data Visualization



01 A PROBLEM WORTH SOLVING

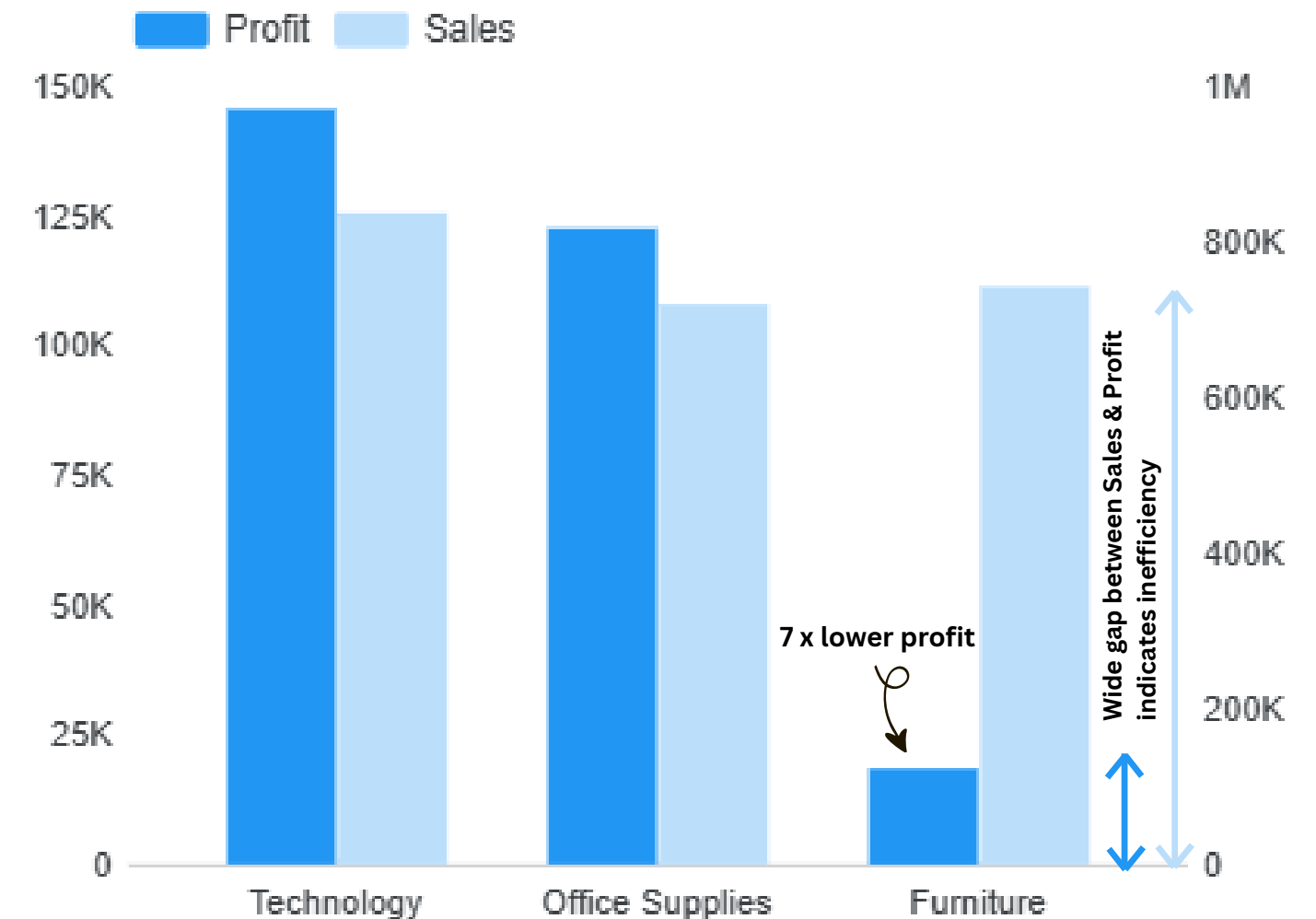
Sales Comparison

Category	Sales (\$)	% of Total Sales
Technology	836,154	~37%
Office Supplies	719,047	~32%
Furniture	741,999	~33%
Total	2,297,200	100%

Profit Comparison

Category	Profit (\$)	% of Total Profit
Technology	145,454	~71%
Office Supplies	122,490	~60%
Furniture	18,451	~9%
Total	286,395	100%

Furniture Generates the Lowest Profit Despite Significant Sales Volume

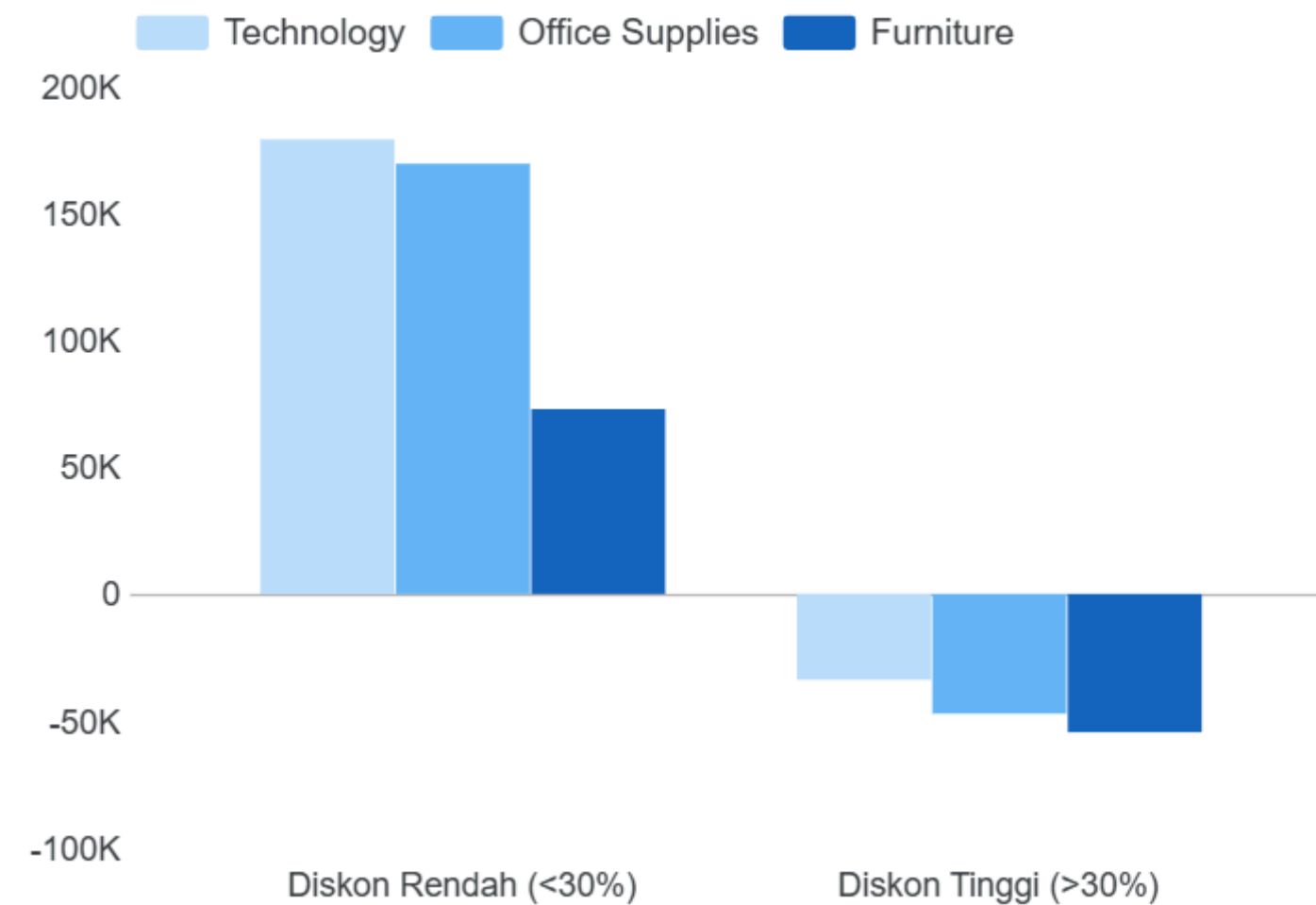


Insight:

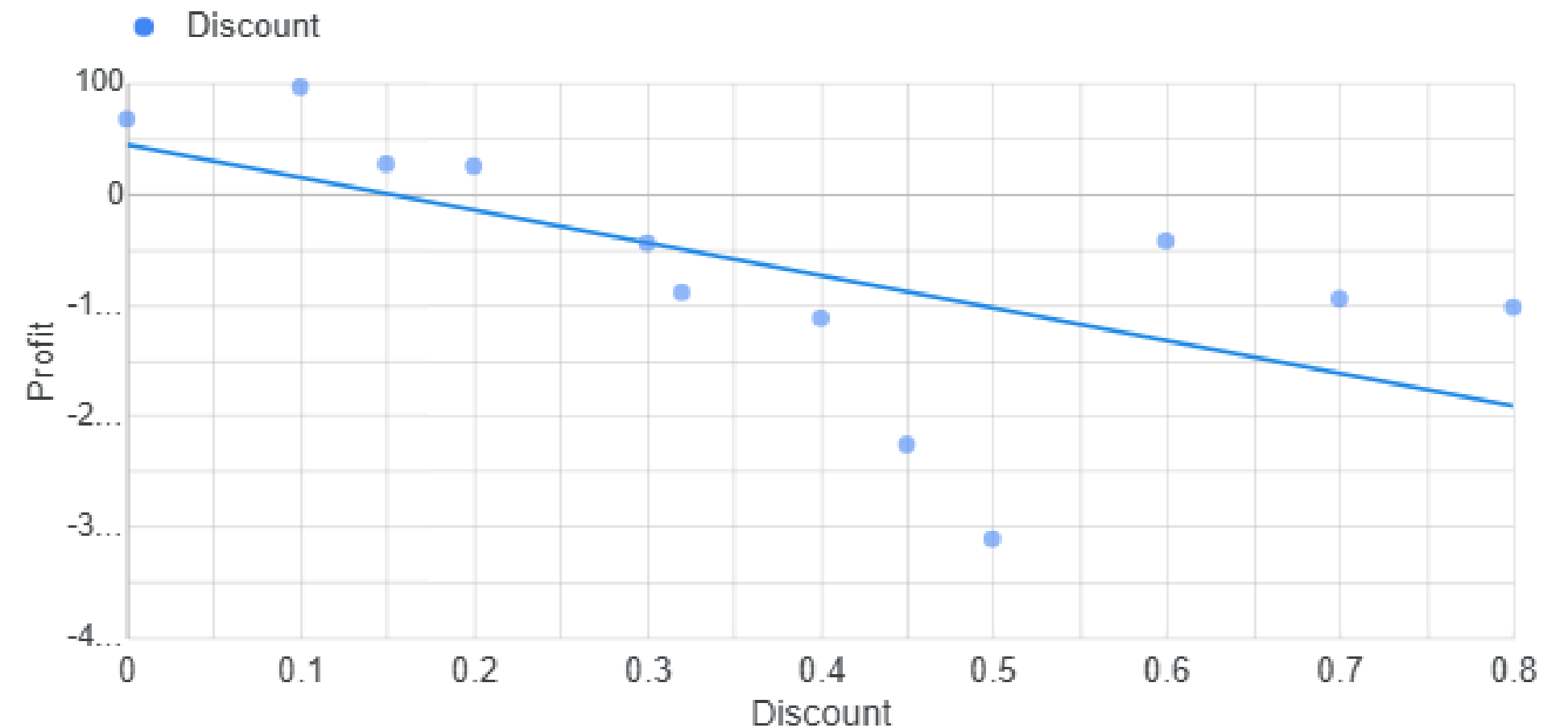
- Furniture contributes **around one-third of total sales**, similar to the other categories, so in terms of sales performance, it's competitive.
- Despite comparable sales, **Furniture contributes only 9% of total profit**, far below Technology and Office Supplies.
- Furniture's **profit margin (2.5%) is nearly 7× lower than** Technology (17.4%) and Office Supplies (17.0%).
- **For every \$1 profit from Furniture**, Technology earns almost \$8, and Office Supplies earns about \$7, even though their sales volumes are similar.

HIGH DISCOUNTS ARE HURTING FURNITURE PROFITABILITY

Profit Drops Sharply When Discounts Exceed 30%, Especially for Furniture

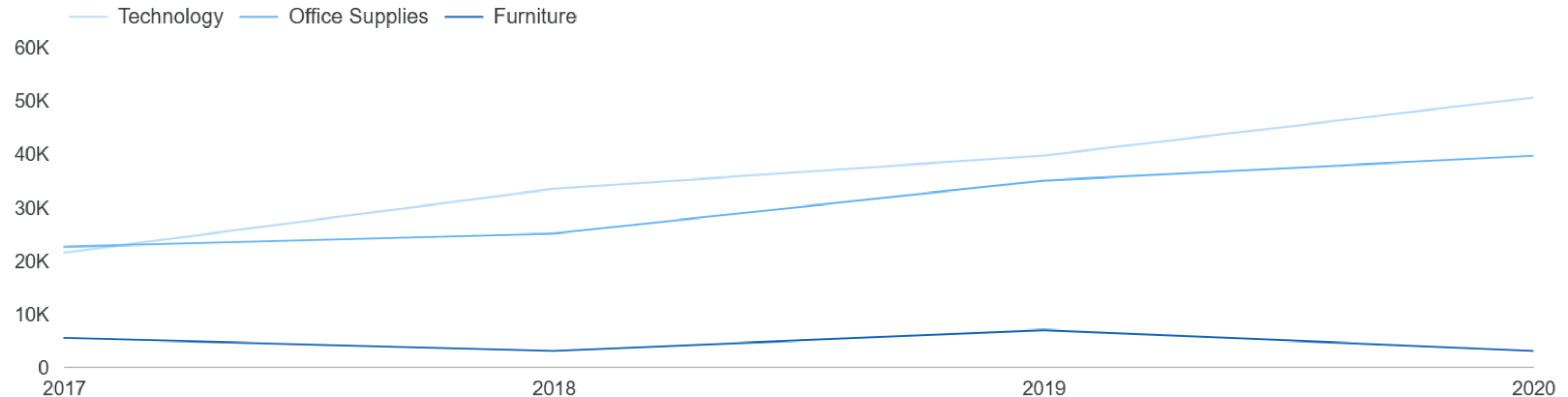


Higher Discounts Lead to Lower Profits



As we can see here, once discounts exceed 30%, profits drop sharply, especially for the Furniture category. The scatter plot confirms a strong negative relationship between discount level and profit. This means that aggressive discounting, particularly in Furniture, is a key reason behind its low profitability.

03 **FURNITURE PROFIT REMAINS FLAT WHILE OTHER CATEGORIES CONTINUE TO GROW**



Technology and Office Supplies show steady improvement in profitability, while Furniture profit remains almost stagnant, indicating persistent inefficiencies in pricing or discount strategies.

This trend suggests that high discounting practices in Furniture have long-term effects, preventing profit from growing despite stable or even increasing sales.

QUANTIFYING THE IMPACT OF DISCOUNT ON PROFIT USING REGRESSION ANALYSIS

- $R^2 = 0.045$, meaning these factors explain about 4.5% of profit variation, significant but indicates other variables (like shipping cost, product type, or region) also influence profit.
- Regression confirms that increasing discounts sharply reduce profit.
- Technology and Office Supplies remain more profitable than Furniture even after controlling for sales and discounts.

Variable	Coefficient (B)	Sig.	Interpretation
(Constant)	233,228.41	<0.001	Baseline profit for <i>Furniture</i> when Sales and Discount = 0
Sales	0.228	<0.001	Each 1 unit increase in Sales raises Profit by 0.228 (positive and significant)
Discount	-1,520,872.596	<0.001	Every 10% increase in discount (e.g., from 20% to 30%) reduces Profit by approximately \$152K — a strong negative and significant effect
Is Office Supplies (1=Yes)	125,104.15	0.004	Office Supplies generate ~125K higher profit than Furniture, controlling for Sales & Discount (significant effect)
Is Technology (1=Yes)	450,250.48	<0.001	Technology yields ~450K higher profit than Furniture (strong and significant effect)

TURNING INSIGHTS INTO ACTION

Key Insight	Business Impact	Recommended Action
High discounts significantly reduce profit (especially in Furniture)	Margin loss up to ~7× lower vs. Technology	Limit discount range; apply data-driven pricing strategy
Furniture underperforms despite high sales volume	Indicates high cost or excessive discounting	Optimize pricing and explore bundle or value-added offers
Technology & Office Supplies yield higher profit margins	Profit concentrated in these categories	Increase marketing investment and inventory allocation for high-margin items



THANK
You!